

JEE MAIN REVISION MASTER: ALTERNATING CURRENT

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 191 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Series LCR Circuit & Impedance	107 PYQs	Circuits
Very High	AC Fundamentals: Peak, RMS & Phase	84 PYQs	Foundations
Very High	Reactance: Inductive & Capacitive	79 PYQs	Impedance Elements
Very High	Resonance, Q-Factor & Bandwidth	54 PYQs	Resonance
Very High	Power in AC Circuits & Power Factor	36 PYQs	Power & Efficiency
Very High	LC Oscillations & Transformers	24 PYQs	Devices & Oscillations
Very High	Alternating Current - Core Application	16 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Series LCR Circuit & Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\phi = \frac{X_L - X_C}{R}$, $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$
AC Fundamentals: Peak, RMS & Phase	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$, $I_{\text{avg, half}} = \frac{2 I_0}{\pi}$
Reactance: Inductive & Capacitive	$X_L = \omega L = 2\pi f L$, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
Resonance, Q-Factor & Bandwidth	$\omega_0 = \frac{1}{\sqrt{LC}}$, $Z_{\min} = R$, $I_{\max} = \frac{V}{R}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$
Power in AC Circuits & Power Factor	$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$, $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$, $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$
LC Oscillations & Transformers	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$, $q(t) = q_0 \cos(\omega_0 t)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Series LCR Circuit & Impedance (Very High Yield — 107 PYQs)

Core Principle: Series LCR impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Current $I = V / Z$. Phase angle $\tan(\phi) = (X_L - X_C) / R$. Voltage across elements $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Caution / JEE Pitfall:** Voltages across series AC components do NOT add algebraically! $V_{\text{total}} \neq V_R + V_L + V_C$. Use phasor vector addition: $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Speed Trick:** If $X_L > X_C$, circuit is inductive (voltage leads current). If $X_C > X_L$, circuit is capacitive (current leads voltage).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." [Ans: C]
- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega, L=1.6\text{H}$ and $C=40\mu\text{F}$ is connected to a variable frequency a.c. source. The inductive reactance at r..." [Ans: 200]

3.2 AC Fundamentals: Peak, RMS & Phase (Very High Yield — 84 PYQs)

Core Principle: Sinusoidal AC quantities: $I_{\text{rms}} = I_0 / \sqrt{2}$ approx $0.707 \cdot I_0$, $V_{\text{rms}} = V_0 / \sqrt{2}$. Average current over full cycle is 0, over half cycle is $(2/\pi) \cdot I_0$ approx $0.637 \cdot I_0$.

■ **Caution / JEE Pitfall:** AC voltmeters and ammeters measure RMS values, not peak values! When JEE specifies '220 V AC', it always denotes V_{rms} .

■ **Speed Trick:** Peak voltage for 220 V AC household line is $220 \cdot \sqrt{2}$ approx 311 V.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Evening Shift] Q#3 (MCQ): "The figure given below shows an LCR series circuit with two switches S_1 and S_2 . When switch S_1 is closed keeping S_2 open, the phase dif..." [Ans: B]
- [2026 Evening Shift] Q#3 (Numerical): "An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rm..." [Ans: 314]

3.3 Reactance: Inductive & Capacitive (Very High Yield — 79 PYQs)

Core Principle: Pure L: voltage leads current by 90° ($\pi/2$), $X_L = \omega \cdot L = 2\pi \cdot f \cdot L$. Pure C: current leads voltage by 90° ($\pi/2$), $X_C = 1 / (\omega \cdot C) = 1 / (2\pi \cdot f \cdot C)$.

■ **Caution / JEE Pitfall:** In DC ($f = 0$), inductor has $X_L = 0$ (short circuit) and capacitor has $X_C = \text{infinity}$ (open circuit/blocks DC).

■ **Speed Trick:** Mnemonic 'CIVIL': In Capacitor (C), Current (I) leads Voltage (V); in Inductor (L), Voltage (V) leads Current (I).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." [Ans: C]
- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega, L=1.6\text{H}$ and $C=40\mu\text{F}$ is connected to a variable frequency a.c. source. The inductive reactance at r..." [Ans: 200]

3.4 Resonance, Q-Factor & Bandwidth (Very High Yield — 54 PYQs)

Core Principle: At resonance: $X_L = X_C$, $\omega_0 = 1 / \sqrt{LC}$, $f_0 = 1 / (2\pi\sqrt{LC})$. Impedance is purely resistive and minimum ($Z = R$), current is maximum ($I = V/R$). $Q = (1/R)\sqrt{L/C} = \omega_0 / \Delta\omega$.

■ **Caution / JEE Pitfall:** At resonance, V_L and V_C are NOT zero! In fact, $V_L = V_C = Q * V_{\text{supply}}$, which can be much larger than supply voltage (voltage magnification)!

■ **Speed Trick:** Bandwidth $\Delta\omega = R / L$. To increase sharpness of resonance (higher Q), decrease R or increase L/C ratio.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega$, $L=1.6H$ and $C=40\mu F$ is connected to a variable frequency a.c. source. The inductive reactance at $r...$ " [Ans: 200]
- [2026 Evening Shift] Q#2 (MCQ): "A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connec..." [Ans: N/A]
- [2026 Morning Shift] Q#4 (Numerical): "Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V=5\sin(100t)$ The values ..." [Ans: 50]

3.5 Power in AC Circuits & Power Factor (Very High Yield — 36 PYQs)

Core Principle: Average power $P_{\text{avg}} = V_{\text{rms}} * I_{\text{rms}} * \cos(\phi)$. Power factor = $\cos(\phi) = R / Z$. In pure L or pure C , $\phi = 90^\circ$, $\cos(\phi) = 0$, so average power dissipated is zero (wattless current $I_{\text{wattless}} = I_{\text{rms}} * \sin(\phi)$).

■ **Caution / JEE Pitfall:** Power factor is $\cos(\phi)$, not $\tan(\phi)$! Always compute Z first: $\cos(\phi) = R / Z$.

■ **Speed Trick:** A choke coil has high inductance L and very low resistance R , allowing it to reduce AC current with negligible power loss ($\cos \phi$ approx 0).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90V$ at frequency $f_{\text{(d)}}=30Hz$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For the series LCR circuit connected with 220V, 50Hz a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." [Ans: C]
- [2026 Morning Shift] Q#2 (Numerical): "An inductor of 10 mH, capacitor of $0.1\mu F$ and a resistor of 100Ω are connected in series across an a.c power supply $\$220V, 70Hz...$ " [Ans: 100]

3.6 LC Oscillations & Transformers (Very High Yield — 24 PYQs)

Core Principle: Ideal transformer: $V_s / V_p = N_s / N_p = I_p / I_s$. Efficiency $\eta = (V_s * I_s) / (V_p * I_p) * 100\%$. LC oscillation frequency $\omega = 1 / \sqrt{LC}$ with energy conservation $U_{\text{total}} = q^2/(2C) + (1/2)*L*i^2$.

■ **Caution / JEE Pitfall:** Transformers work ONLY with AC, never with steady DC ($d\phi/dt = 0$ for DC, so secondary voltage is zero).

■ **Speed Trick:** In step-up transformer ($N_s > N_p$), voltage increases but current decreases, keeping total power constant.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connec..." [Ans: N/A]
- [2024 Morning Shift] Q#1 (MCQ): "In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to..." [Ans: C]
- [2024 Morning Shift] Q#2 (MCQ): "Primary coil of a transformer is connected to 220V ac. Primary and secondary turns of the transforms are 100 and 10 respectively. Secondar..." [Ans: N/A]

3.7 Alternating Current - Core Application (Very High Yield — 16 PYQs)

Core Principle: Key problem pattern in Alternating Current.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- **[2025 Evening Shift] Q#3 (MCQ):** "In the circuit shown here, assuming threshold voltage of diode is negligibly small, then voltage $V_{(AB)}$ is correctly represented by:..." **[Ans: N/A]**
- **[2023 Evening Shift] Q#3 (Numerical):** "A square shaped coil of area 70cm^2 having 600 turns rotates in a magnetic field of 0.4wbm^{-2} , about an axis which is parallel to o..." **[Ans: 44]**
- **[2023 Evening Shift] Q#3 (Numerical):** "In an ac generator, a rectangular coil of 100 turns each having area $14 \times 10^{-2}\text{m}^2$ is rotated at 360rev/min about an axis perpe..." **[Ans: 1584]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A LCR series circuit driven with $E_{(\text{rms})}=90\text{V}$ at frequency $f_{(d)}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactance $X_{(L)}=20.0\Omega$ and capacitance with capacitive reactance $X_{(C)}=80.0\Omega$. The power factor of the circuit is ____.

- (A) 0.8 **[Correct]**
- (B) 0.64
- (C) 0.9
- (D) 0.5

Correct Answer: A

Solution Analysis: In an AC circuit, the power factor ($\cos\phi$) is defined as the ratio of the resistance (R) to the total impedance (Z) of the circuit : $\text{Power Factor} = \cos\phi = (R / Z)$ For a series LCR circuit, the impedance is $Z = \sqrt{R^2 + (X_{(C)} - X_{(L)})^2}$ The value of resistance is $R=80\Omega$, the inductive reactance is $X_{(L)}=20.0\Omega$, and the capacitive reactance is $X_{(C)}=80.0\Omega$. $X_{(\text{net})} = |X_{(L)} - X_{(C)}| = |20 - 80| = 60\Omega$ So, the impedance is, $Z = \sqrt{(80^2) + 60^2} = \sqrt{(6400 + 3600)} = \sqrt{(10000)} = 100\Omega$ Thus, the power factor is, $\cos\phi = \frac{R}{Z} = \frac{80}{100}$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value of α is ____.

- (A) 4
- (B) 8
- (C) 6 **[Correct]**
- (D) 10

Correct Answer: C

Solution Analysis: The power factor ($\cos\phi$) of an AC circuit is defined as the ratio of the true resistance (R) to the total impedance (Z) of the circuit. $\text{Power Factor} = \cos\phi = (R / z)$ In a series LCR circuit, the inductive reactance ($X_{(L)}$) and capacitive reactance ($X_{(C)}$) oppose each other. The net reactance is the difference between them. $X = |X_{(L)} - X_{(C)}|$ Substituting the given values : $X = |70 - 150|\Omega = 80\Omega$ The total impedance of the circuit is, $Z = \sqrt{R^2 + X^2}$ $Z = \sqrt{(60^2) + (80^2)} = \sqrt{(10000)} = 100\Omega$ So, the power factor is $\cos\phi = \frac{R}{Z} = \frac{60}{100} = \frac{6}{10}$

Benchmark Problem 3: JEE Main 2025 (Online) 22nd January Evening Shift (Q#1)

A series LCR circuit is connected to an alternating source of emf E . The current amplitude at resonant frequency is I_0 . If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

- (A) $\frac{I_0}{2}$ **[Correct]**
- (B) $\frac{I_0}{\sqrt{2}}$
- (C) $2I_0$
- (D) I_0

Correct Answer: A

Solution Analysis: At resonance in a series LCR circuit, the reactive components (inductive and capacitive) cancel each other out, leaving only the resistance. Therefore, the amplitude of the current is given by: $I_0 = (E / R)$ Now, if the resistance is doubled, the new resistance becomes $2R$. The current amplitude at resonance then becomes: $I = (E / 2R) = (1 / 2)((E / R)) = \frac{I_0}{2}$ Thus, the amplitude of current at resonance when the resistance is doubled is $\frac{I_0}{2}$. The correct option is A.